

ZHICHAO “CARTON” ZENG

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PROFESSIONAL EXPERIENCE

Texas A&M University

Postdoctoral Associate

– Advisor: Louis Strigari

College Station, Texas, USA

9/2024-present

EDUCATION

The Ohio State University

Ph.D. in Physics

- Thesis: Evolution and properties of small substructures in self-interacting dark matter cosmology
- Advisor: Annika H. G. Peter

Columbus, Ohio, U.S.

8/2018 – 8/2024

The Chinese University of Hong Kong

M.Phil. in Physics

- Thesis: Effects of cosmological neutrinos on the large-scale structure formation
- Advisor: Ming-Chung Chu

Hong Kong

8/2016 – 6/2018

The Chinese University of Hong Kong

B.Sc. in Physics

- Thesis: Influence of massive and degenerate neutrinos on the evolution of the Universe
- Advisor: Ming-Chung Chu

Hong Kong

9/2012 – 6/2016

RESEARCH INTERESTS

N-body Simulation: Small and large-scale dark matter structure, (sub)halo dynamics

Alternative DM Models: Self-interacting dark matter (SIDM)

Observational Probes: Substructure lensing, dwarf galaxies, stellar streams

PUBLICATIONS & PRE-PRINTS

Total Publications (14)

Total Citations: 428

Leading Author Publications (5)

Total Citations: 158

- [1] **Z. C. Zeng**, O. Hartl, L. Strigari, A. Peter, X. Du, C. Mace, A. Benson, *Bypassed Core Formation in Milky Way-Mass SIDM Halos: Implications for the Local Group Past-Pericenter Scenario*, 2026, arXiv:2604.08647. submitted to *Phys.Rev.D* cited by: 0
- [2] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, J. Li, et al., *Diversity and universality: evolution of dwarf galaxies with self-interacting dark matter*, 2025, *Phys.Rev.D*, arXiv:2412.14621. cited by: 10
- [3] **Z. C. Zeng**, A. Peter, X. Du, S. Yang, A. Benson, et al., *Till the core collapses: the evolution and properties of self-interacting dark matter subhalos*, 2025, *Phys.Rev.D*, arXiv:2310.09910. cited by: 27
- [4] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, S. Kim, et al., *Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter subhalo*, 2022, *MNRAS*, arXiv:2110.00259. cited by: 101
- [5] **Z. C. Zeng**, S. Yeung and M.-C. Chu, *Effects of neutrino mass and asymmetry on cosmological structure formation*, 2019, *JCAP*, arXiv:1808.00357. cited by: 20

Mentored Student Publications (3)

Total Citations: 53

- [1] C. Mace, B. Dhanasingham, **Z. C. Zeng**, et al., *The Sensitivity of Substructure Lensing to SIDM Core-collapse Model Variation*, 2026, *Phys.Rev.D*, arXiv:2605.24174. cited by: 2

- [2] C. Mace, S. Yang, **Z. C. Zeng**, et al., *Calibrating the SIDM Gravothermal Catastrophe with N-body Simulations*, 2025, Phys.Rev.D, arXiv:2504.13004. *cited by: 11*
- [3] C. Mace, **Z. C. Zeng**, A. Peter, et al., *Convergence Tests of Self-Interacting Dark Matter Simulations*, 2024, Phys.Rev.D, arXiv:2402.01604. *cited by: 40*

Collaborative Publications (5)

Total Citations: 217

- [1] A. Engelhardt, et al. (inc. **Z. C. Zeng**), *MARVELously Dark: the gravothermal evolution of dwarf halos in velocity-dependent SIDM*, 2026, arxiv:2601.23264. *submitted to JCAP* *cited by: 5*
- [2] X. Du, A. Benson, **Z. C. Zeng**, et al., *Tidal evolution of cored and cuspy dark matter halos*, 2024, Phys.Rev.D, arXiv:2403.09597. *cited by: 32*
- [3] S. Yang, et al. (inc. **Z. C. Zeng**), *A quantitative comparison between velocity dependent SIDM cross-sections constrained by the gravothermal and isothermal models*, 2024, MNRAS, arXiv:2305.05067. *cited by: 26*
- [4] S. Yang, X. Du, **Z. C. Zeng**, A. Benson, F. Jiang, E. O. Nadler, A. Peter, *Gravothermal Solutions of SIDM Halos: Mapping from Constant to Velocity-dependent Cross Section*, 2023, ApJ, arXiv:2205.02957. *cited by: 79*
- [5] F. Jiang, et al. (inc. **Z. C. Zeng**), *A semi-analytic study of self-interacting dark-matter haloes with baryons*, 2023, MNRAS, arXiv:2206.12425. *cited by: 57*
- [6] R. Emami, et al. (inc. **Z. C. Zeng**), *Evidence of Neutrino Enhanced Clustering in a Complete Sample of Sloan Survey Clusters Implying $\sum m_\nu = 0.119 \pm 0.034$ eV*, 2017, arXiv:1711.05210. *cited by: 18*

PRESENTATIONS

Self-interacting dark matter subhalos and dwarf satellite galaxies

- Poster, Galactic Frontiers II, Dartmouth college June 2025
- Poster, Gravity in the Local Group, CMU June 2025
- Plenary talk, Theoretical Astroparticle and Cosmology Symposium in Texas, Texas A&M Oct 2024
- Texas A&M Astro Seminar Sept 2024
- Plenary talk, Dark Matter 2024, KITP, UCSB Jun 2024
- Ohio State HEP Seminar Mar 2024

Till the core collapses: evolution of SIDM subhalos

- Plenary Talk, Galactic Frontiers, CCA Flatiron Institute July 2023
- Seminar (Virtual), Nature of Dark Matter, Yale June 2023

Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter

- Plenary talk (virtual), N-body-shop workshop, CCA Flatiron Institute June 2022
- Seminar (virtual), The McWilliams Center for Cosmology, CMU Apr 2022
- Poster (virtual), COSMO21 July 2021

MENTORING

Steven Zhu (undergraduate Student at OSU & UMich)	2025 – Present
Odelia Hartl (Graduate Student at TAMU)	2024 – Present
Chloe Zheng (Graduate Student at OSU)	2024 – Present
Angelica Whisnant (Graduate Student at OSU)	2024 – Present
Charles Mace (Graduate Student at OSU)	2021 – Present
Jiashen He (Undergraduate Student at OSU)	2019 summer

TEACHING

The Ohio State University	2019 – 2022
<i>Graduate Teaching Assistant</i>	
– Courses: Mechanics, Fluids, Waves (Introductory Physics).	
The Chinese University of Hong Kong	2016 – 2018
<i>Graduate Teaching Assistant</i>	
– Courses: Thermodynamics, Computational Physics, General Physics.	

TEACHING EXPERIENCE

Graduate Teaching Assistant

2018, 2019, 2022

The Ohio State University

Columbus, OH

- Courses: Mechanics, Fluids, Waves (Introductory Physics for Engineers and Physicists).
- Led recitation sections and supervised laboratory experiments for undergraduate students.

Graduate Teaching Assistant

2016 – 2018

The Chinese University of Hong Kong

Hong Kong

- Courses: Thermodynamics, Computational Physics, General Physics.
- Designed computational physics assignments and provided technical support for numerical simulations.

AWARDS & HONORS

OSU Presidential Fellowship (Gradschool Dissertation Year Fellowship, \$36k+tuition)

2023 – 2024

Professor Charles K. Kao Research Exchange Scholarship (\$5k)

2013 – 2014

University Full Admission Scholarship (CUHK, \$25k+tuition)

2012 – 2016

SKILLS

Programming: Python, C, Bash script

Scientific Codes: Gadget2, Arepo, Amiga Halo Finder, Glamer

Languages: Mandarin (Native), Cantonese (Proficient), English (Proficient)